

Wei Chen

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Education

University of Wisconsin–Madison

Aug 2024 – Dec 2025

M.S. in Electrical and Computer Engineering

Major: Machine Learning and Signal Processing (MLSP)

GPA: 3.73 / 4.00

Sichuan University

Sep 2020 – Jun 2024

B.S. in Flight Vehicle Control and Information Engineering

GPA: 3.79 / 4.00

Relevant Coursework

Large Language Models and Deep Learning

- Advanced Topics in Large Language Models (ECE 901)
- Introduction to Artificial Neural Networks (CS 539)

Machine Learning and Optimization

- Machine Learning (ECE 760)
- Matrix Methods in Machine Learning (ECE 532)
- Topics in Mathematical Data Science (ECE 888)

Perception and Signal Processing

- Image Processing (ECE 533)

Theoretical Foundations(Mostly Mastered, with Ongoing Review and Exploration in Some Areas)

Classical ML, DL, and Mathematical Foundations

- ERM-based learning: logistic regression, SVM, regularization, generalization (bias–variance trade-off)
- Neural networks: MLP, CNN, RNN/LSTM; BackPropagation, Nonlinear activations, Normalization, and Gradient-based optimization
- Mathematical foundations: SVD, QR, EigenDecomposition, Least Squares, Low-Rank approximation, MLE/MAP, Cross-Entropy, KL divergence

Large Language Models

- **Architectures**
 - Model paradigms: encoder–decoder (e.g., T5), encoder-only (e.g., BERT), decoder-only (e.g., GPT)
 - Efficient scaling: Mixture-of-Experts (MoE), DeepSeekMoE
 - Alternative sequence models: Mamba (state space models), LLaDA
 - Advanced training objectives: Multi-Token Prediction (MTP)
 - Positional encoding: RoPE, LongRoPE, YaRN
 - Theoretical insights: scaling laws, data–compute trade-offs
 - Optimization perspective: training dynamics, stability, and convergence
- **Training Data (Data-Centric LLMs)**
 - Scaling laws for data vs model size (e.g., Chinchilla)
 - Dataset construction pipelines: DataComp-LM
 - Data curation strategies:
 - * Filtering (quality control)
 - * Deduplication
 - * Data mixing (multi-domain balancing)
 - * Curriculum learning (ordering data for training)

- **Training Algorithms (Optimization)**
 - Adaptive optimizers: Adam, AdamW
 - Emerging optimizers: Lion, Shampoo, SOAP, Muon
- **Post-Training / Alignment**
 - Supervised Fine-Tuning (SFT)
 - Reinforcement Learning from Human Feedback (RLHF)
 - Preference alignment and safety tuning
- **Using LLMs (Inference & Systems)**
 - Reasoning frameworks: ReAct
 - Tool-use: Toolformer
 - Programmatic prompting: DSPy, GEPA
 - Efficiency:
 - * FlashAttention
 - * KV cache optimization (e.g., SnapKV)
 - * Quantization
- **Adapting LLMs**
 - (1) **Parameter-Efficient Fine-Tuning (PEFT)**
 - **LoRA**: Introduces low-rank adapters to update weights efficiently with minimal additional parameters and latency.
 - **DoRA**: Decomposes weight updates into magnitude and direction, improving LoRA’s expressiveness without increasing inference cost.
 - **Expressive Power of LoRA**: Theoretical analysis of when low-rank updates can approximate full fine-tuning in Transformers.
 - **Convergence of LoRA**: Provides guarantees and identifies failure modes under practical training regimes.
 - **QLoRA (optional)**: Combines LoRA with quantization for memory-efficient fine-tuning.
 - (2) **In-Context Learning (ICL)**
 - **Z-ICL**: Constructs pseudo-demonstrations from raw text to bridge zero-shot and few-shot performance.
 - **Dual Operating Modes of ICL**: Interprets ICL as either task retrieval or implicit learning within the context.
 - **Prompt Intern**: Internalizes prompts to reduce token usage and inference cost.
 - (3) **Continual Adaptation & Prompt Evolution**
 - **Self-Evolving Agents Survey**: Framework for autonomous improvement via iterative feedback and adaptation.
 - **PromptBreeder**: Evolves prompts through mutation and selection mechanisms.
 - **Auto Evol-Instruct**: Automatically generates and refines instruction data without human supervision.
 - **Automatic Prompt Engineer (APE)**: Formulates prompt optimization as program search.
 - **PromptAgent**: Uses planning and reflection to iteratively refine prompts toward expert-level performance.